Flight Test Engineering Performance Data Analysis

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Flight Overview

Airframe: Cessna 182

Takeoff & departure from KPDK

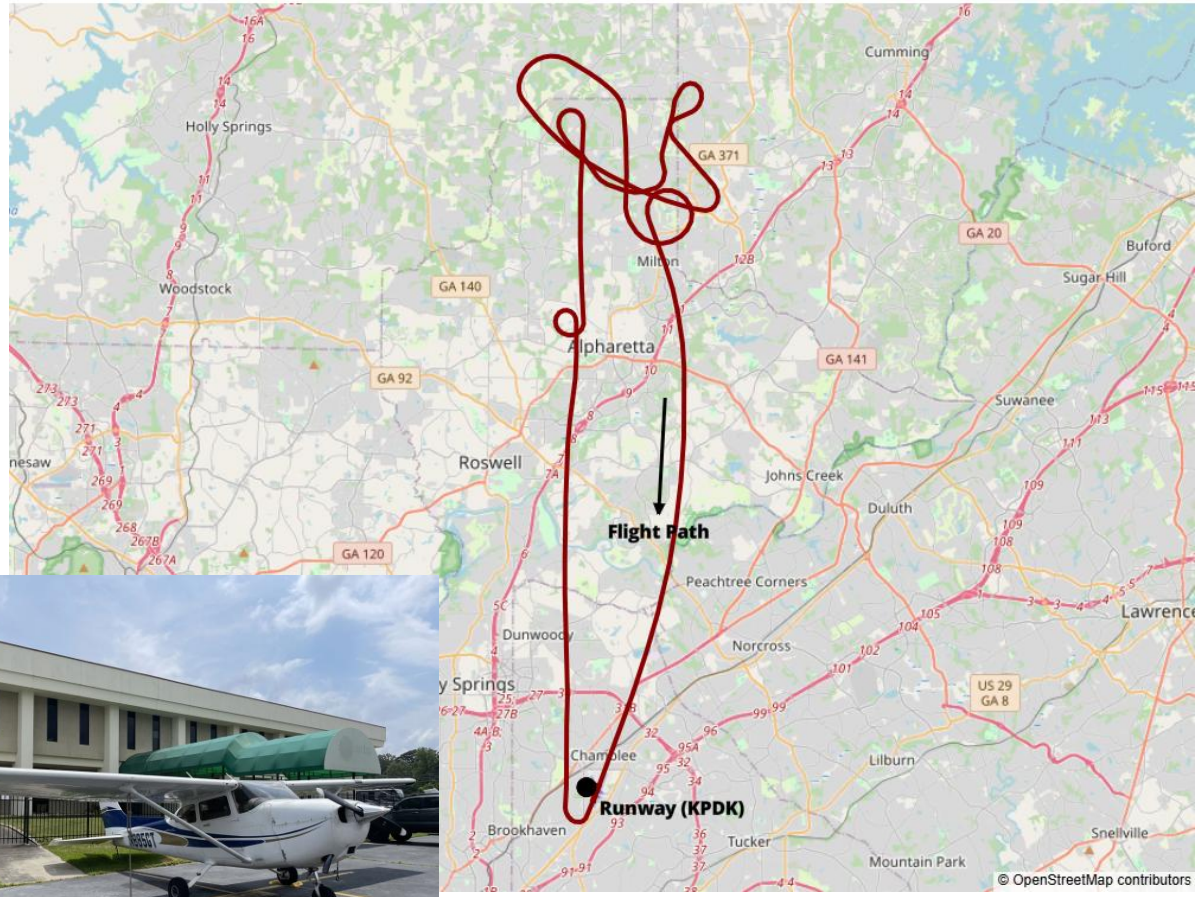
- Flight time from ~14:30 → ~15:15 EST
  - Feb 25, 2024 (OAT 18 °C)
- Runway altitude 1000 ft MSL

Cruise @ 5K ft MSL, 100-120 KIAS

- Turns (15° – 30°) performed at cruise altitude
- Stall maneuver (reference speed 56 KIAS)

Performance data obtained via onboard measurement unit

Flight Path (GPS Imagery)



Similar Cessna 172 operated by YJFC: https://yjfc.org/our-fleet.html

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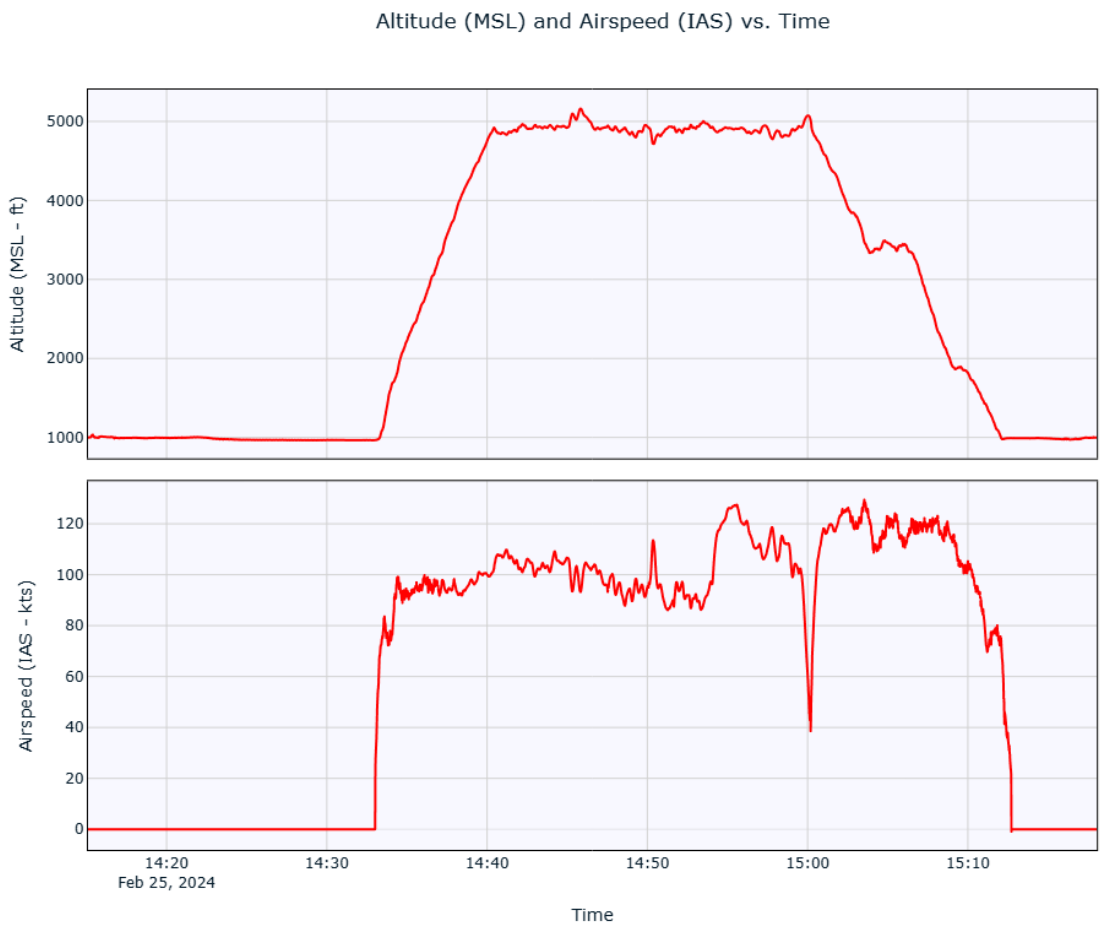
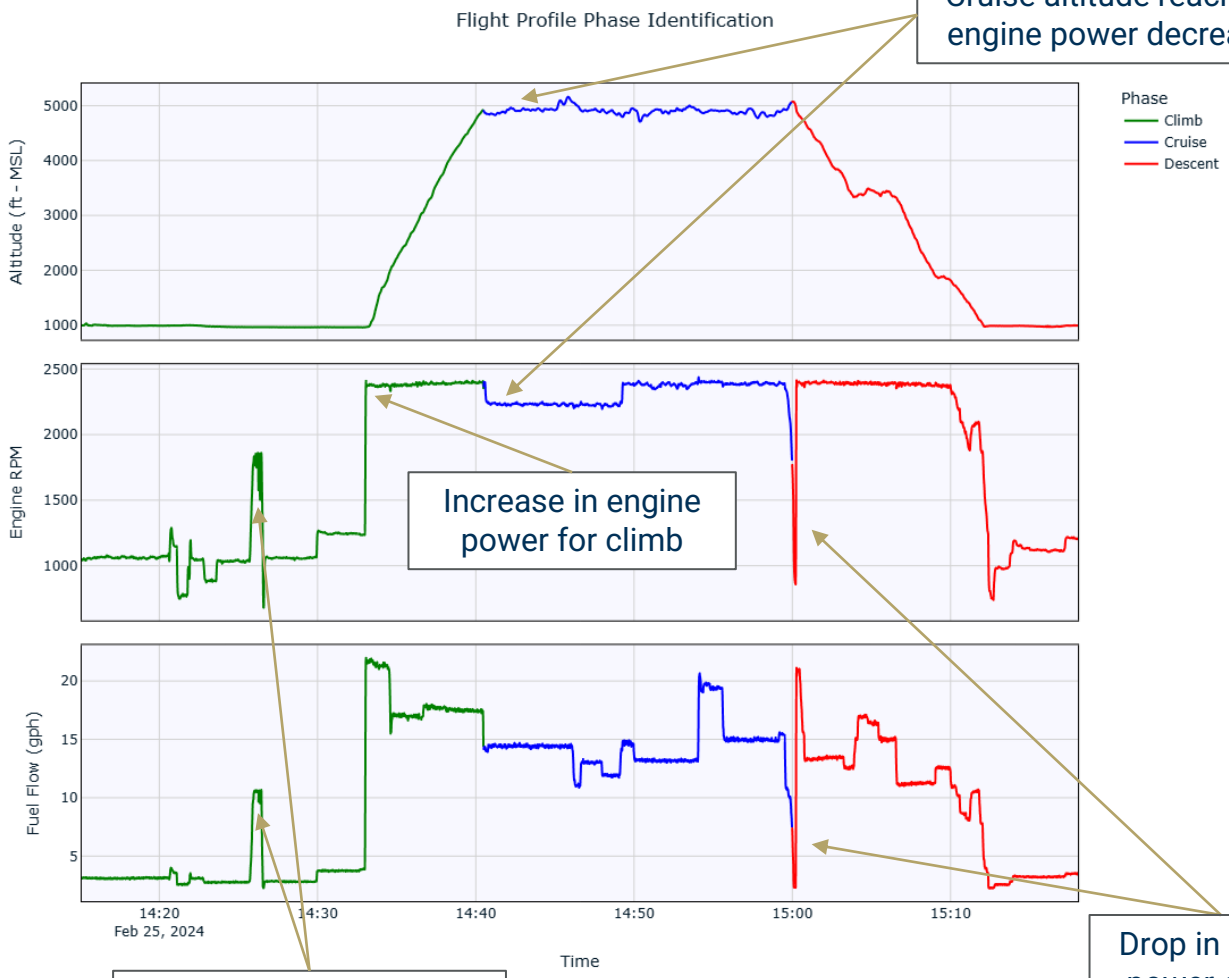
Profile Analysis



Profile Analysis

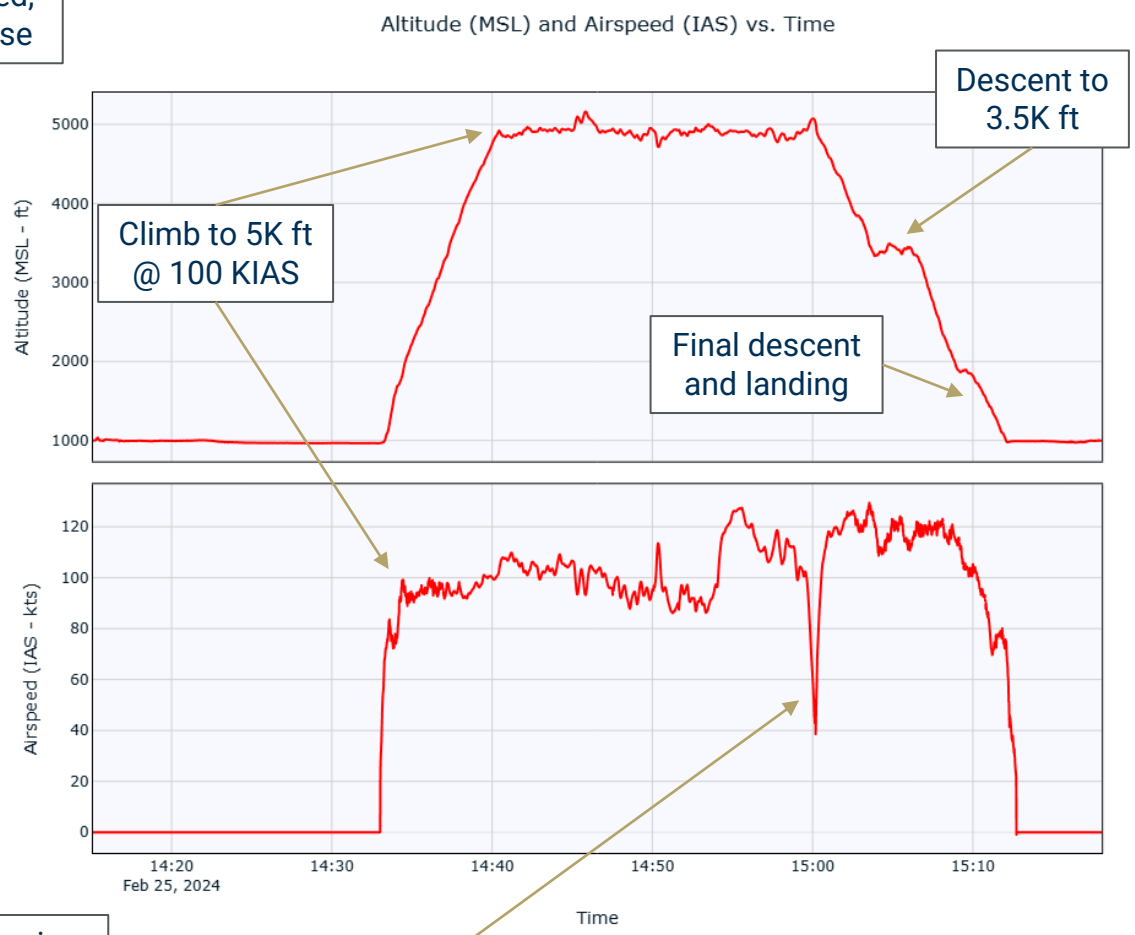


Cruise altitude reached, engine power decrease

Increase in engine power for climb

Max power preflight test

Drop in engine power output



Descent to 3.5K ft

Climb to 5K ft @ 100 KIAS

Final descent and landing

Stall airspeed passed

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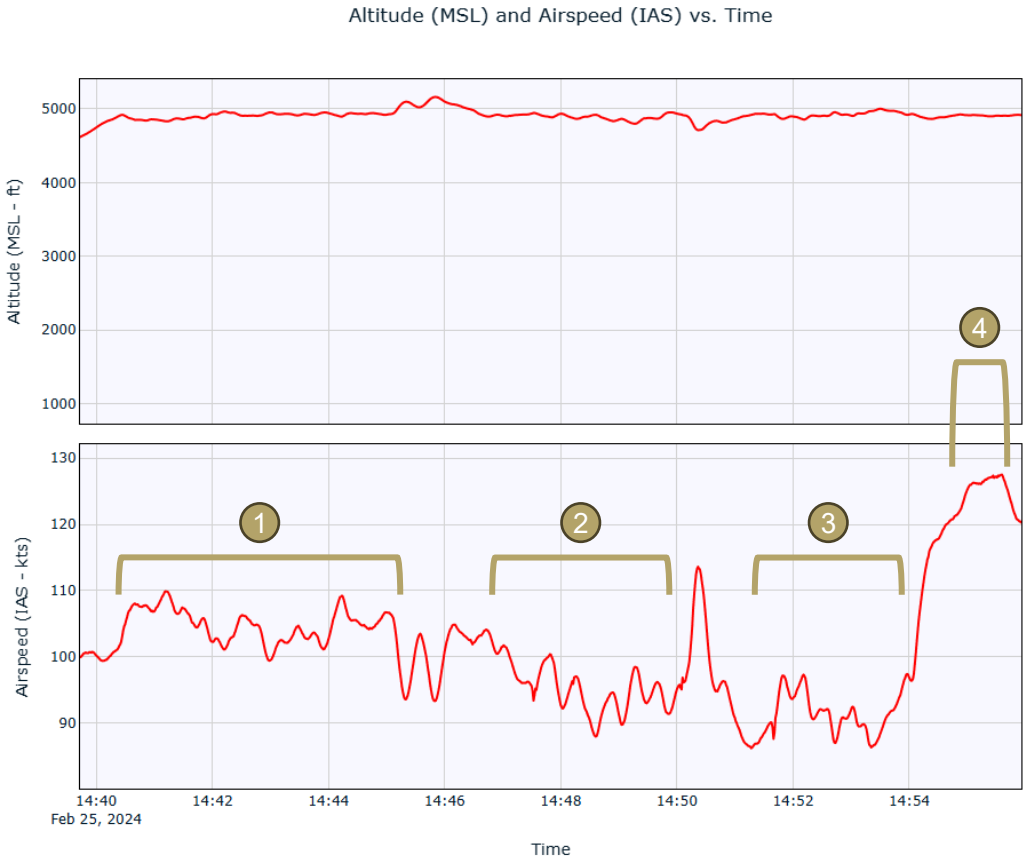
Stable Cruise Segments

	Start Time	End Time	Duration	Avg Alt (ft)	Avg IAS (kt)	Avg TAS (kt)	Avg RPM	Fuel Flow (gph)	OAT (°C)
1	14:40:30	14:45:15	285s	4919	104.8	112.8	2238	14.4	7.1
2	14:46:48	14:49:48	180s	4896	95.4	102.6	2256	13	7
3	14:51:38	14:53:54	136s	4934	91.6	98.6	2380	13.2	7.2
4	14:54:40	14:55:59	79s	4919	124.3	133.9	2397	18.2	6.9

Stable cruise segments defined by variation ≤ 30 ft with period > 60 s



Inertial Analysis

Two turns @ 30° bank (left and right)

Multiple 15° – 20° turns (left and right)

A/C maintained positive pitch throughout most of flight

Stall event at ~15:00 EST

- Sudden pitch down (-10°)
- Spike in measured inertial load factor (+1.4 G – 0.65 G)
- Fluctuation in roll angle during stall recovery (analysis continued …)

** Load factor can be verified with the approximation $n = \frac{1}{cos\phi}$, where ϕ is the bank angle and n is the load factor.*

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Stall Event

Stall occurs < 56 KIAS in Cessna 182

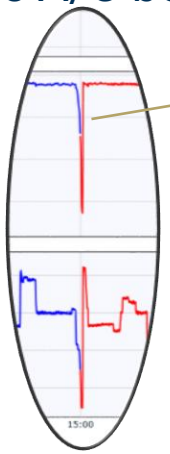
- Stall during flight visible in graphs
  - Sharp drop in pitch attitude
  - Drop and recovery in IAS

Airspeed recovery due to pilot intervention

- Intentional power cut and recovery to induce stall

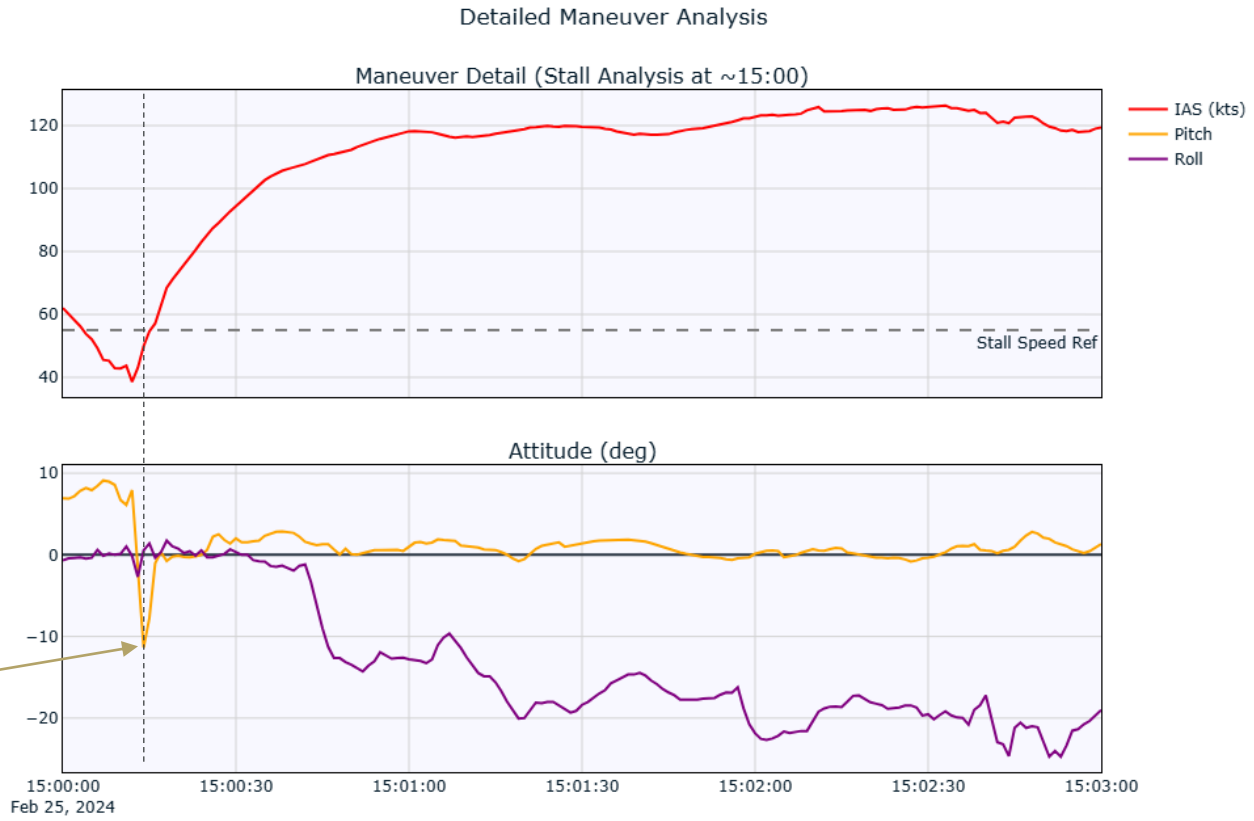
Roll angle unstable shortly after stall

- Roll angle from ~15:45:00 onward is to change the heading of the A/C before landing



Engine RPM vs Time

Fuel Flow vs Time



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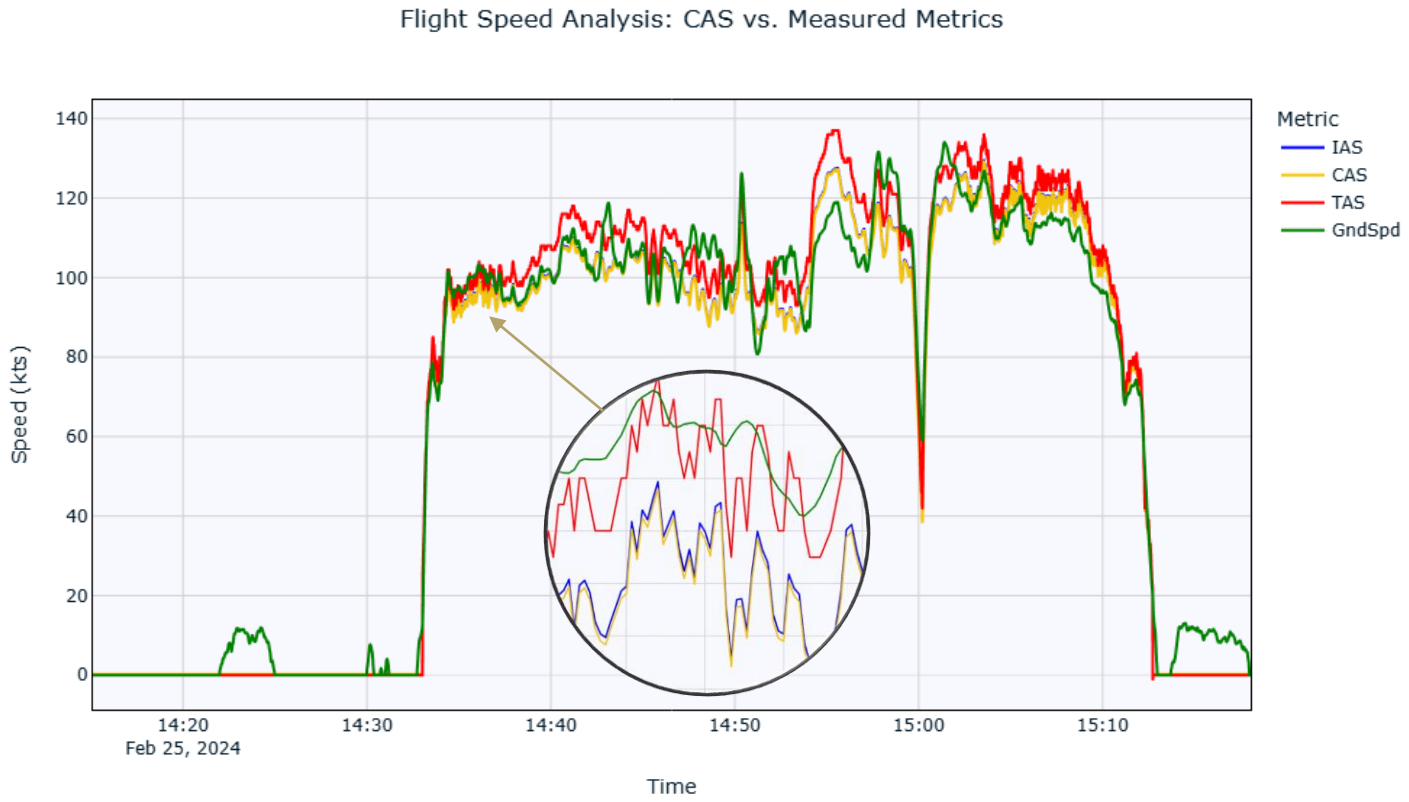
Airspeed Corrections

CAS closely aligns with IAS, given by Cessna 182T POH, but can be approximated with a 2nd order polynomial fitted to the data:

$$CAS = 0.000043(IAS)^2 + 0.9906(IAS) + 0.2263$$



CESSNA
MODEL 182T NAV III

SECTION 5
PERFORMANCE

AIRSPEED CALIBRATION

NORMAL STATIC SOURCE

CONDITIONS:

Power required for level flight or maximum power descent.

FLAPS UP												
KIAS	55	60	70	80	90	100	110	120	130	140	150	160
KCAS	61	65	73	82	90	99	108	117	126	136	146	156
FLAPS 20°												
KIAS	40	50	60	70	80	90	100	110	120	---	---	---
KCAS	52	57	64	72	80	88	98	108	118	---	---	---
FLAPS FULL												
KIAS	40	50	60	70	80	90	95	---	---	---	---	---
KCAS	51	56	63	71	80	89	94	---	---	---	---	---

C182 POH Airspeed Calibration Data [1]

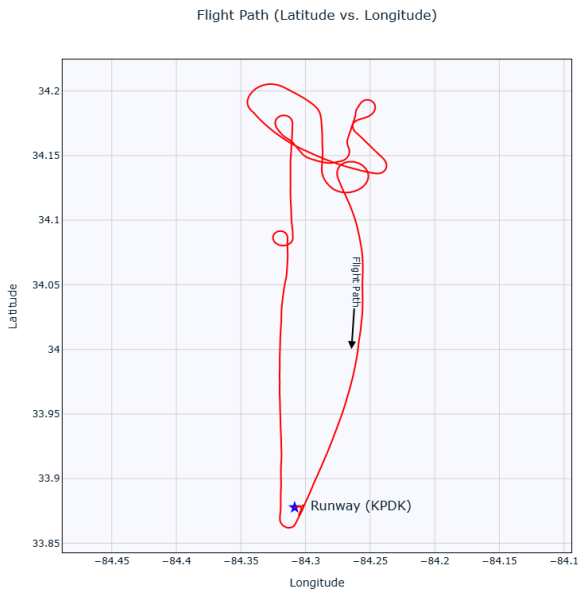
Recap

40-minute flight from KPDK

Testing altitude b/w 4,500 and 5,000 ft MSL

- Turn and stall performance evaluated

Flight duration (total)	1:03:00
Flight duration (in air)	0:40:00
Max speed	130 KIAS
Min speed (stall)	39 KIAS
Max altitude	5,169 ft MSL
Max RPM	2,439 RPM
Avg fuel flow (flight)	14.4 gph
Max +G	1.48



KPDK Airport:
exploregeorgia.org/atlanta/transportation-rentals/airport/peachtree-dekalb-airport

References

[1] https://txwg.cap.gov/media/cms/N780CP_1FE2B1E59AAFF.pdf

- Graphs generated with plotly (Python) -
- Gemini used for script development (code submitted separately) -

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